

MONTHLY DEPARTMENTAL PROGRESS REPORT

Department of Electrical & Electronics Engineering

Reporting Period: April 2026 • Date of Submission: 25/04/2026

REPORTING OVERVIEW & DEPARTMENT PROFILE

Department Name	Department of Electrical & Electronics Engineering
Head of Department (HOD)	Mr. K. Gangadhar
Reporting Cycle Period	April 2026
Date of Submission	25/04/2026
Activities Summary	35 recorded activities • 4 Publications • 4 Syllabus Course(s)

1A. JOURNAL PUBLICATIONS (2)

01 Design of Compact Triple-Band Reconfigurable MIMO Antenna for Sub-6 GHz 5G and V2X Communications

Author(s): Dr. V. Annapurna, Er. K. Ramesh • Journal: IEEE Antennas and Wireless Propagation Letters

ISSN/ISBN: 1536-1225 Issue: 23/4 Pages: 780-784 Indexed in: IEEE / SCI / Scopus

Paper Link / DOI: <https://doi.org/10.1109/LAWP.2026.3190284>**02 Low-Power High-Throughput Approximate Multiplier Architecture for Edge AI Vision Processors in 28nm CMOS**

Author(s): Dr. V. Annapurna, Er. B. Madhavi • Journal: Integration, the VLSI Journal (Elsevier)

ISSN/ISBN: 0167-9260 Issue: 88 Pages: 102-114 Indexed in: Elsevier / Scopus

Paper Link / DOI: <https://doi.org/10.1016/j.vlsi.2026.102114>

1B. CONFERENCE PRESENTATIONS (2)

01 FPGA Implementation of Hardware-Accelerated Convolutional Filter for Real-Time Biomedical Signal Denoising

Author(s): Dr. V. Annapurna, T. Akhila • Conference: International Conference on VLSI Design (VLSID 2026)

Date: 10th April 2026 Location/Mode: Bengaluru, India Indexing: IEEE Xplore

Proceedings Link: <https://vlsiconference.org/papers/ece-441>**02 Beamforming Optimization in Millimeter-Wave Massive MIMO Networks Using Deep Reinforcement Learning**

Author(s): Er. K. Ramesh, Dr. V. Annapurna • Conference: IEEE International Conference on Communications & Signal Processing (ICCSP 2026)

Date: 21st April 2026 Location/Mode: Chennai, India (Hybrid) Indexing: IEEE Xplore

Proceedings Link: <https://iccsp2026.org/proceedings/ece-92>

1C. PATENTS (2)

01 Wearable Multi-Frequency Textile Antenna Patch for Continuous Non-Invasive Physiological Monitoring

Inventors: Dr. V. Annapurna, Er. B. Madhavi • Applicants: Sanskriti School of Engineering

Patent No: 202641031209 A | Status: Published | Date: 09/04/2026

02 Low-Latency Neuromorphic Sensory Interface Device for Autonomous Agricultural Robotics

Inventors: Dr. V. Annapurna, Dr. Kethineni Vinod Kumar • Applicants: Sanskriti School of Engineering

Patent No: 202541077182 B | Status: Awarded / Granted | Date: 01/04/2026

1D. ENTREPRENEURSHIP & START-UP INITIATIVES (2)

- 01 Hardware Hackathon: Drone Avionics & Embedded IoT Sensing for Precision Agriculture**
Date: 09/04/2026 | Participants: 75
Key Outcomes: 3 hardware prototypes constructed; 1 team shortlisted for T-Hub hardware grant
- 02 Start-up Pitch on Solar-Powered LoRaWAN Nodes for Rural Water Tank Monitoring**
Date: 19/04/2026 | Participants: 32
Key Outcomes: Prototype successfully field-tested in Beedupalli water reservoir

2. NSS & OTHER EXTENSION ACTIVITIES (2)

- 01 Solar Lantern Assembly & Clean Green Energy Distribution Drive for Rural Households**
Date: 13/04/2026 | Venue: Nagireddypalli Tribal Hamlet | Participants: 48 Volunteers
Outcomes: Assembled and distributed 40 solar emergency LED lanterns to families without continuous grid power
- 02 Electronic Waste Disposal Awareness & Free Home Appliance Repair Camp**
Date: 25/04/2026 | Venue: Kothacheruvu Community Center | Participants: 35 Student Technicians
Outcomes: Repaired 52 non-functional domestic electronic appliances free of cost for local villagers

3. FACULTY DEVELOPMENT PROGRAMS (FDPS) (2)

- 01 One-Week National Level FDP on RISC-V Architecture & Open-Source Silicon Design Flow**
Organizing Body: IIT Madras & SSE Department of ECE • **Dates:** 14/04/2026 to 19/04/2026
Role: Convener & Participant (Hybrid Mode) | **Key Outcomes:** Faculty gained expertise in OpenLane ASIC toolchain, Verilog verification, and tape-out readiness
- 02 AICTE ATAL FDP on 5G/6G Wireless Physical Layer Design and SDR Prototyping**
Organizing Body: NIT Calicut • **Dates:** 06/04/2026 to 10/04/2026
Role: Participant (Online Mode) | **Key Outcomes:** Hands-on simulation using USRP Software Defined Radios and GNU Radio toolkits

4. STUDENT DEVELOPMENT PROGRAMS (SDPS) (2)

- 01 Hands-on Skill Certification on ARM Cortex-M4 Embedded C & FreeRTOS Architecture**
Date: 16/04/2026 to 18/04/2026 • **Resource Person:** Er. N. Bharath (Senior Firmware Architect, Qualcomm India) • **Participants:** 65
Outcomes: 65 students developed real-time task scheduling and peripheral driver modules
- 02 Masterclass on Multilayer High-Speed PCB Layout Design using Altium Designer**
Date: 22/04/2026 • **Resource Person:** Er. T. Manoj (Senior Hardware Designer, Bosch India) • **Participants:** 50
Outcomes: 50 students routed 4-layer impedance-matched boards for RF transceiver circuits

5. HONORS, AWARDS & CERTIFICATIONS

Faculty Honor: Dr. V. Annapurna — IETE Best Research Paper Gold Medal in RF & Microwave Engineering (Institution of Electronics and Telecommunication Engineers (IETE), 17/04/2026)

Faculty Honor: Er. B. Madhavi — Outstanding Woman Engineer in VLSI Innovation Award (IEEE Women in Engineering (WIE) - Hyderabad Section, 07/04/2026)

Student Honor: P. Sai Teja (22SSE1A0441) & M. Bhavana (22SSE1A0419) — 1st Prize & Rs. 50,000 Cash in Texas Instruments India Innovation Design Challenge at TI National Innovation Challenge 2026 (Texas Instruments India & AICTE, 11/04/2026 to 12/04/2026)

Student Honor: K. Sneha (23SSE1A0467) — 2nd Prize in National Level Embedded Circuit Bug-Hunt Competition at ELECTROKRITHI 2026 (NIT Rourkela, 19/04/2026)

Certification: VLSI System on Chip (SoC) Design & Verification (NPTEL-AICTE FDP Certification) on SWAYAM / NPTEL (IIT Kharagpur) — Certified: 41 Certified (14 with Elite Silver)

Certification: Embedded Systems & Industrial IoT Architecture (Industry Credential) on Texas Instruments Education Online — Certified: 46 Certified

Certification: Digital Signal Processing with MATLAB & Simulink (MathWorks Certified Specialist) on MathWorks Training Portal — Certified: 33 Certified

6A. DEPARTMENT MEETINGS & ACADEMIC GOVERNANCE (2)

01 Department Meeting #1

Date: 02.04.2026

Decisions & Key Discussions: Review of Texas Instruments Innovation Lab procurement, major project evaluation rubrics, and NAAC Criterion 2 documentation.

Policy Updates: Mandatory hardware working prototype demonstration required for final year mini-projects.

Minutes Link: <https://sseptp.org/iqac/ece/minutes-02-04-2026>

02 Department Meeting #2

Date: 18.04.2026

Decisions & Key Discussions: Mid-term evaluation results review, syllabus completion tracking, and scheduling of ELECTROFEST 2026 technical association events.

Policy Updates: Allocation of extra 2 hours of lab time weekly for students working on competitive hardware challenges.

Minutes Link: <https://sseptp.org/iqac/ece/minutes-18-04-2026>

6B. COLLABORATIONS & MOUS (2)

1. Texas Instruments University Program & SSE Department of ECE • Period: Valid 2024 to 2027 (3 Years) • Scope: Establishment of Advanced MCU & Embedded IoT Lab, Industrial Training, and Student Hardware Project Funding

2. Cadence Design Systems Academic Network • Period: Valid 2025 to 2028 (3 Years) • Scope: VLSI EDA Software Licenses, Custom IC Design Flow Training, and Student Placement Enablement

7-9. TECHNICAL ASSOCIATIONS & INNOVATION INITIATIVES

Additional Initiative: Open Hardware Innovation & IoT Tinkering Hub (05/04/2026) — Established a 24/7 maker space equipped with digital oscilloscopes, soldering stations, and 3D printer for electronics hobbyists.

Additional Initiative: Peer-to-Peer Circuit Debugging & PCB Soldering Clinic (15/04/2026) — Weekly clinic where senior students mentor juniors on SMD soldering techniques and breadboard troubleshooting.

Technical Association: ELECTROKRITHI 2026 - National Level Technical Symposium & Project Expo (21/04/2026) — Conducted circuit debugging challenge, technical paper presentations, and robotics track race.

Technical Association: Guest Lecture on Next-Generation 6G Wireless Networks & Terahertz Communication (11/04/2026) — Gained insights into reconfigurable intelligent surfaces (RIS) and satellite ground station antennas.

IIC Cell: Idea Pitch on Low-Power IoT Environmental Warning Systems for Flash Floods (15/04/2026) — Under IIC 6.0, students demonstrated ultrasonic water level telemetry devices for rural causeways.

IIC Cell: Hands-on Patent Drafting Clinic for Embedded Electronics & Chip Architecture (26/04/2026) — Practical session on drafting circuit claims, block diagrams, and hardware patent disclosures.

10. SYLLABUS COVERAGE & CURRICULUM PROGRESS (4)

S.No	Subject / Course	Year/Sem	Faculty	% Done	% Pend	Remarks
1	Digital Signal Processing & Applications	III B.Tech II Sem	Dr. V. Annapurna	100%	Nil	Completed. FFT algorithms, digital filter design, and MATLAB simulations verified.
2	VLSI Design & Technology	III B.Tech II Sem	Er. B. Madhavi	95%	5%	CMOS subsystem design and stick diagrams concluding; revision scheduled.
3	Microprocessors and Microcontrollers (8086 / ARM)	II B.Tech II Sem	Er. K. Ramesh	92%	8%	ARM interrupt handling and timer configuration concluding this week.
4	Antennas and Wave Propagation	III B.Tech II Sem	Er. G. Prasad	100%	Nil	Completed. Microstrip patch antenna and array design question banks shared.

Mr. K. Gangadhar
Head of Department

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